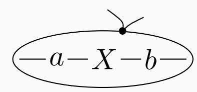
Round 2: every approach targets the book's *current* notation

All approaches now share the book's real `tikz-styles.tex` (`triangle`, `copydot`, `\dloop`, `\dwhiskers`). Test: eq. (70)-style derivative row with the Penrose loop.

A. Legs as node options

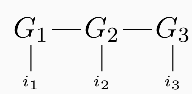
A1: derivative of $a^T Xb$ (Penrose loop)

```
\begin{tikzpicture}[tn]
  \node[legs={180}] (a) {$a$};
  \node[right=.7em of a] (X) {$X$};
  \node[right=.7em of X, legs={0}] (b) {
      $b$};
  \draw (a) -- (X) -- (b);
  \dloop[55]{(a)(b)}
  \dwhiskers
\end{tikzpicture}
```



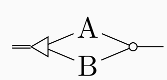
A2: TT chain

```
\begin{tikzpicture}[tn]
  \node[legs={270/i_1}] (G1) {$G_1$};
  \node[right=1em of G1, legs={270/i_2}] (
      G2) {$G_2$};
  \node[right=1em of G2, legs={270/i_3}] (
      G3) {$G_3$};
  \draw (G1) -- (G2) -- (G3);
\end{tikzpicture}
```



A3: Khatri–Rao $A * B$ (book geometry)

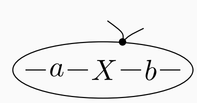
```
\begin{tikzpicture}[tn]
  \node[triangle, rotate=180] (f) {};
  \node (A) at (.6,.25)  {A};
  \node (B) at (.6,-.25) {B};
  \node[copydot, legs={0}] (c) at (1.2,0)
      {};
  \draw (f) -- (A) -- (c);
  \draw (f) -- (B) -- (c);
  \draw[double] (f) -- ++(-.4,0);
\end{tikzpicture}
```



B. The graphs library

B1: chain + loop drawn after

```
\begin{tikzpicture}[tn]
\graph[grow right sep=.7em] {
  l[as=, coordinate] -- a[as=$a$] --
  X[as=$X$] -- b[as=$b$] --
  r[as=, coordinate]
};
\dloop[55]{(a)(b)}
\dwhiskers
\end{tikzpicture}
```

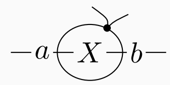


B3: Khatri–Rao fan — does not work. Two attempts (`branch down sep`, explicit `yshifts`) could not make the `graphs` placement engine center a fan-out/fan-in between two triangles; the branch nodes land on or below the chain axis and downstream nodes misalign. Verdict: approach B is pleasant for *chains* (B1 composes fine with `\dloop`), but branching shapes should use approach A or D.

C. Mini-DSL (scope: inline chains)

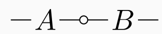
C1: derivative row — loop via trailing ’

```
\tn{-a-{X’}-b-}
```



C2: Hadamard-style chain

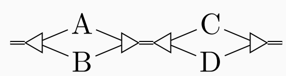
```
\tn{-A-*-B-}
```



D. Pics (molecules)

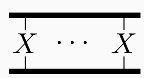
D2: mixed product (511) by composition

```
\begin{tikzpicture}[tn]
  \pic (k1) {kron={A}{B}};
  \pic (k2) at (1.7,0) {kron={C}{D}};
  \draw[double] (k1-in) -- ++(-.35,0);
  \draw[double] (k1-out) -- (k2-in);
  \draw[double] (k2-out) -- ++(.35,0);
\end{tikzpicture}
```



D3: determinant stack

```
\begin{tikzpicture}[tn]
  \pic {det stack=X};
\end{tikzpicture}
```

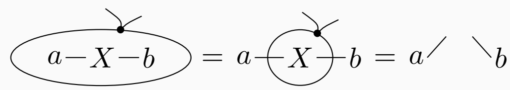


E. Paper reproduction test (approach A + `\dloop`)

The forms the book actually uses, including subset loops.

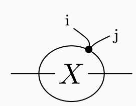
R1: eq (70) full derivation — whole loop, X-only loop, result

```
\begin{tikzpicture}[tn]
  \node (a) {$a$};
  \node[right=.7em of a] (X) {$X$};
  \node[right=.7em of X] (b) {$b$};
  \draw (a) -- (X) -- (b);
  \dloop[55]{(a)(b)}
  \dwhiskers
\end{tikzpicture}
$=$
\begin{tikzpicture}[tn]
  \node (a) {$a$};
  \node[right=1em of a] (X) {$X$};
  \node[right=1em of X] (b) {$b$};
  \draw (a) -- (X) -- (b);
  \dloop[55]{(X)}
  \dwhiskers
\end{tikzpicture}
$=$
\begin{tikzpicture}[tn]
  \node (a) {$a$};
  \node[right=2.2em of a] (b) {$b$};
  \draw (a.east) -- ++(.25,.28);
  \draw (b.west) -- ++(-.25,.28);
\end{tikzpicture}
```



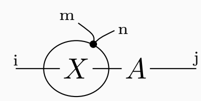
R2: eq (73) — lone X, labeled whiskers, free edges through the loop

```
\begin{tikzpicture}[tn]
  \node (X) {$X$};
  \draw (X) -- ++(-.8,0);
  \draw (X) -- ++(.8,0);
  \dloop[55]{(X)}
  \draw (ddot) .. controls +(80:.12) ..
      ++(-.2,.28)
     node[pos=1, anchor=south east, font=\
         tiny] {i};
  \draw (ddot) .. controls +(45:.12) ..
      ++(.28,.18)
     node[pos=1, anchor=west, font=\tiny]
         {j};
\end{tikzpicture}
```



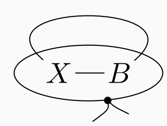
R3: eq (74) — X-only loop inside a longer chain

```
\begin{tikzpicture}[tn]
  \node (X) {$X$};
  \node[right=1em of X] (A) {$A$};
  \draw (X) -- ++(-.8,0)
     node[pos=1, above, font=\tiny] {i};
  \draw (X) -- (A);
  \draw (A) -- ++(.8,0)
     node[pos=1, above, font=\tiny] {j};
  \dloop[55]{(X)}
  \draw (ddot) .. controls +(80:.12) ..
     ++(-.2,.28)
     node[pos=1, anchor=south east, font=\
         tiny] {m};
  \draw (ddot) .. controls +(45:.12) ..
     ++(.28,.18)
     node[pos=1, anchor=west, font=\tiny]
         {n};
\end{tikzpicture}
```



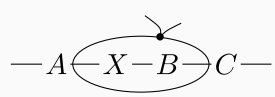
R4: trace row (108) — arc crossing the loop, dot below

```
\begin{tikzpicture}[tn]
  \node (X) {$X$};
  \node[right=1em of X] (B) {$B$};
  \draw (X) -- (B);
  \path (X) edge[out=140, in=40, looseness
     =2.6] (B);
  \dloop[-55]{(X)(B)}
  \dwhiskersdown
\end{tikzpicture}
```



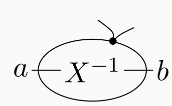
R5: subset loop spanning two nodes mid-chain

```
\begin{tikzpicture}[tn]
  \node (A) {$A$};
  \node[right=1em of A] (X) {$X$};
  \node[right=.7em of X] (B) {$B$};
  \node[right=1em of B] (C) {$C$};
  \draw (A) -- (X) -- (B) -- (C);
  \draw (A) -- ++(-.6,0);
  \draw (C) -- ++(.6,0);
  \dloop[55]{(X)(B)}
  \dwhiskers
\end{tikzpicture}
```



R6: splitting-rule mid-step — loop around an inverse

```
\begin{tikzpicture}[tn]
  \node (a) {$a$};
  \node[right=1em of a] (X) {$X^{-1}$};
  \node[right=1em of X] (b) {$b$};
  \draw (a) -- (X) -- (b);
  \dloop[55]{(X)}
  \dwhiskers
\end{tikzpicture}
```



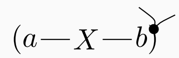
F. Before/after gallery (old code verbatim from the book)

G1. Derivative row (70) LHS

new = if Penrose loops are adopted

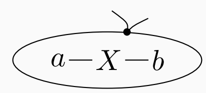
OLD (9 lines)

```
\begin{tikzpicture}[baseline=(a0.base),
    inner sep=1pt]
   \node (a0) {$(a$};
   \node[right=1em of a0] (a1) {$X$};
   \node[right=1em of a1] (a2) {$b)$};
   \path (a0) edge (a1);
   \path (a1) edge (a2);
   \draw[d] (a2.east) .. controls
      +(80:.12) .. ++(-.2,.28);
   \draw[d] (a2.east) .. controls
      +(45:.12) .. ++(.28,.18);
\end{tikzpicture}
```



NEW (8 lines)

```
\begin{tikzpicture}[tn, baseline=(a0.base)
    ]
   \node (a0) {$a$};
   \node[right=.8em of a0] (a1) {$X$};
   \node[right=.8em of a1] (a2) {$b$};
   \draw (a0) -- (a1) -- (a2);
   \dloop[55]{(a0)(a2)}
   \dwhiskers
\end{tikzpicture}
```

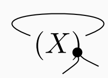


G2. Trace row (108), middle step

loop + down whiskers

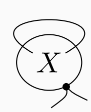
OLD (6 lines)

```
\begin{tikzpicture}[baseline=(a1.base),
    inner sep=1pt]
   \node (a1) {$(X)$};
   \path (a1) edge[out=160, in=20, loop,
      looseness=4] (a1);
   \draw[d] (a1.south east) .. controls
      +(280:.12) .. ++(-.2,-.28);
   \draw[d] (a1.south east) .. controls
      +(315:.12) .. ++(.28,-.18);
\end{tikzpicture}
```



NEW (6 lines)

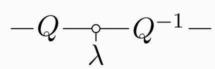
```
\begin{tikzpicture}[tn, baseline=(a1.base)
    ]
   \node (a1) {$X$};
   \path (a1) edge[out=150, in=30, loop,
      looseness=7] (a1);
   \dloop[-55]{(a1)}
   \dwhiskersdown
\end{tikzpicture}
```



G3. Eigendecomposition (1.1)

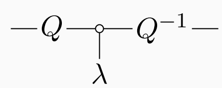
OLD (11 lines)

```
\begin{tikzpicture}[baseline=(Q.base),
   inner sep=1pt]
\node (Q) {$Q$};
\draw (Q.west) -- ++(-.3, 0);
\node[right=1em of Q, copydot] (dot) {};
\path (Q) edge (dot);
\node[below=.3em of dot] (e) {$\lambda$};
\path (dot) edge (e);
\node[right=1em of dot] (Qi) {$Q^{-1}$};
\path (dot) edge (Qi);
\draw (Qi.east) -- ++(.3, 0);
\end{tikzpicture}
```



NEW (7 lines)

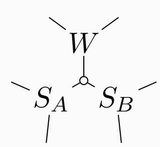
```
\begin{tikzpicture}[tn, baseline=(Q.base),
    every leg label/.style={}]
  \node[legs={180}] (Q) {$Q$};
  \node[right=1em of Q, copydot,
        legs={270/\lambda}] (dot) {};
  \node[right=1em of dot, legs={0}] (Qi) {
     $Q^{-1}$};
  \draw (Q) -- (dot) -- (Qi);
\end{tikzpicture}
```



G4. Strassen star

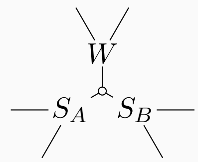
OLD (19 lines)

```
\begin{tikzpicture}[inner sep=1pt,
   baseline=.0em]
    \node[copydot] (m) at (0,0) {};
    \node (W) at (0,.5) {$W$};
    \node (SB) at (-30:.5) {$S_B$};
    \node (SA) at (-150:.5) {$S_A$};
    \node (w1) at (60:1) {};
    \node (w2) at (120:1) {};
    \node (sa1) at (-120:1) {};
    \node (sa2) at (-180:1) {};
    \node (sb1) at (-0:1) {};
    \node (sb2) at (-60:1) {};
    \draw (m) -- (W);
    \draw (m) -- (SA);
    \draw (m) -- (SB);
    \draw (W) -- (w1) {};
    \draw (W) -- (w2) {};
    \draw (SB) -- (sb1) {};
    \draw (SB) -- (sb2) {};
    \draw (SA) -- (sa1) {};
    \draw (SA) -- (sa2) {};
\end{tikzpicture}
```



NEW (7 lines)

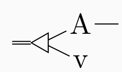
```
\begin{tikzpicture}[tn, baseline=0em, leg
   length=.5]
  \node[copydot] (m) at (0,0) {};
  \node[legs={60,120}]    (W)  at (0,.5)
       {$W$};
  \node[legs={-120,180}] (SA) at (-150:.5)
     {$S_A$};
  \node[legs={0,-60}]     (SB) at (-30:.5)
     {$S_B$};
  \draw (m) -- (W)  (m) -- (SA)  (m) -- (
     SB);
\end{tikzpicture}
```



G5. Kronecker-vector product $A \otimes v$

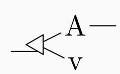
OLD (9 lines)

```
\begin{tikzpicture}[baseline=-.25em, inner
    sep=1pt]
   \node[triangle, rotate=180] (F0) at
      (0,0) {};
   \node (A) at (.5,.25) {A};
   \node (B) at (.5,-.25) {v};
   \draw[double] (F0) -- ++(-.4,0);
   \draw (F0) -- (A) -- ++(.5, 0);
   \draw (F0) -- (B);
\end{tikzpicture}
```



NEW (6 lines)

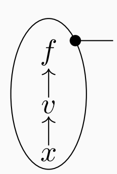
```
\begin{tikzpicture}[tn]
   \node[triangle, rotate=180, legs={180},
         every leg/.style={double}] (F0)
            {};
   \node[legs={0}] (A) at (.5,.25) {A};
   \node (v) at (.5,-.25) {v};
   \draw (F0) -- (A)  (F0) -- (v);
\end{tikzpicture}
```



G6. Chain rule (functions ch.) — manual ellipse vs auto-fit

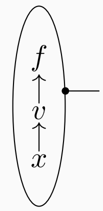
OLD (10 lines; hand-tuned center and radii)

```
\begin{tikzpicture}[baseline=-1em, inner
   sep=1pt]
   \node (n0) {$f$};
   \node[below=1em of n0] (n1) {$v$};
   \node[below=1em of n1] (n2) {$x$};
   \draw[->] (n2) -- (n1);
   \draw[->] (n1) -- (n0);
   \drawellipse{0}{-.55}{.5}{1}{180/4}
\end{tikzpicture}
```



NEW (8 lines; ellipse fits itself)

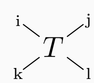
```
\begin{tikzpicture}[tn, baseline=-1em]
   \node (n0) {$f$};
   \node[below=1em of n0] (n1) {$v$};
   \node[below=1em of n1] (n2) {$x$};
   \draw[->] (n2) -- (n1);
   \draw[->] (n1) -- (n0);
   \dloop[30]{(n0)(n1)(n2)}
   \draw (ddot) -- ++(.45,0);
\end{tikzpicture}
```



G7. Order-4 tensor with labeled legs

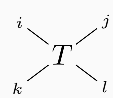
OLD style (9 lines)

```
\begin{tikzpicture}[baseline=-.25em, inner
     sep=1pt]
   \node (T) {$T$};
   \draw (T) -- ++(-.4,.3)
      node[pos=1.3, font=\tiny] {i};
   \draw (T) -- ++(.4,.3)
      node[pos=1.3, font=\tiny] {j};
   \draw (T) -- ++(-.4,-.3)
      node[pos=1.3, font=\tiny] {k};
   \draw (T) -- ++(.4,-.3)
      node[pos=1.3, font=\tiny] {l};
\end{tikzpicture}
```



NEW (3 lines)

```
\begin{tikzpicture}[tn]
  \node[legs={143/i, 37/j, 217/k, -37/l}]
      (T) {$T$};
\end{tikzpicture}
```



G8. Where the new approach does NOT win

Kronecker sandwich (511): approach A saves only the two outer stub lines; the real compression is approach D's `kron` pic (16 $\to$ 7, shown in D2 above). **Bent identity wires (511b), twists (510), wrapped traces:** plain TikZ is already minimal — a library adds nothing. `\vecmatvec` / `\trace` / `\detstack` **rows:** the existing macros already beat everything; they stay.